

Wandering Black Hole Feeds on Its Gravitational Wake

A 35,000-solar-mass intermediate-mass black hole accretes from a dense downstream wake, revealing a new growth channel.

arXiv:2608.10719 — Multiwavelength evidence for off-nuclear IMBH accretion via gravitational focusing.

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Background

Intermediate-mass black holes are considered seeds of supermassive black holes, but many remain displaced from galactic nuclei after hierarchical assembly. Off-nuclear black holes lack access to conventional fueling channels, leaving their growth uncertain.

Direct observational evidence for gravitational-wake accretion onto wandering intermediate-mass black holes has remained elusive.

What this paper is selling

This paper reports a 35,000-solar-mass wandering black hole accreting through a gravitational wake. Multi-epoch spectroscopy reveals distinct upstream, downstream, and absorber components driving rapidly changing-look variability.

Key results to remember

- Broad lines, compact continuum, variability, and power-law SED confirm the black hole nature.
- Three gas components: blueshifted upstream flow, redshifted dense wake, and optically thick absorbers.
- Absorbers within the capture radius drive rapidly changing-look broad-line variability.

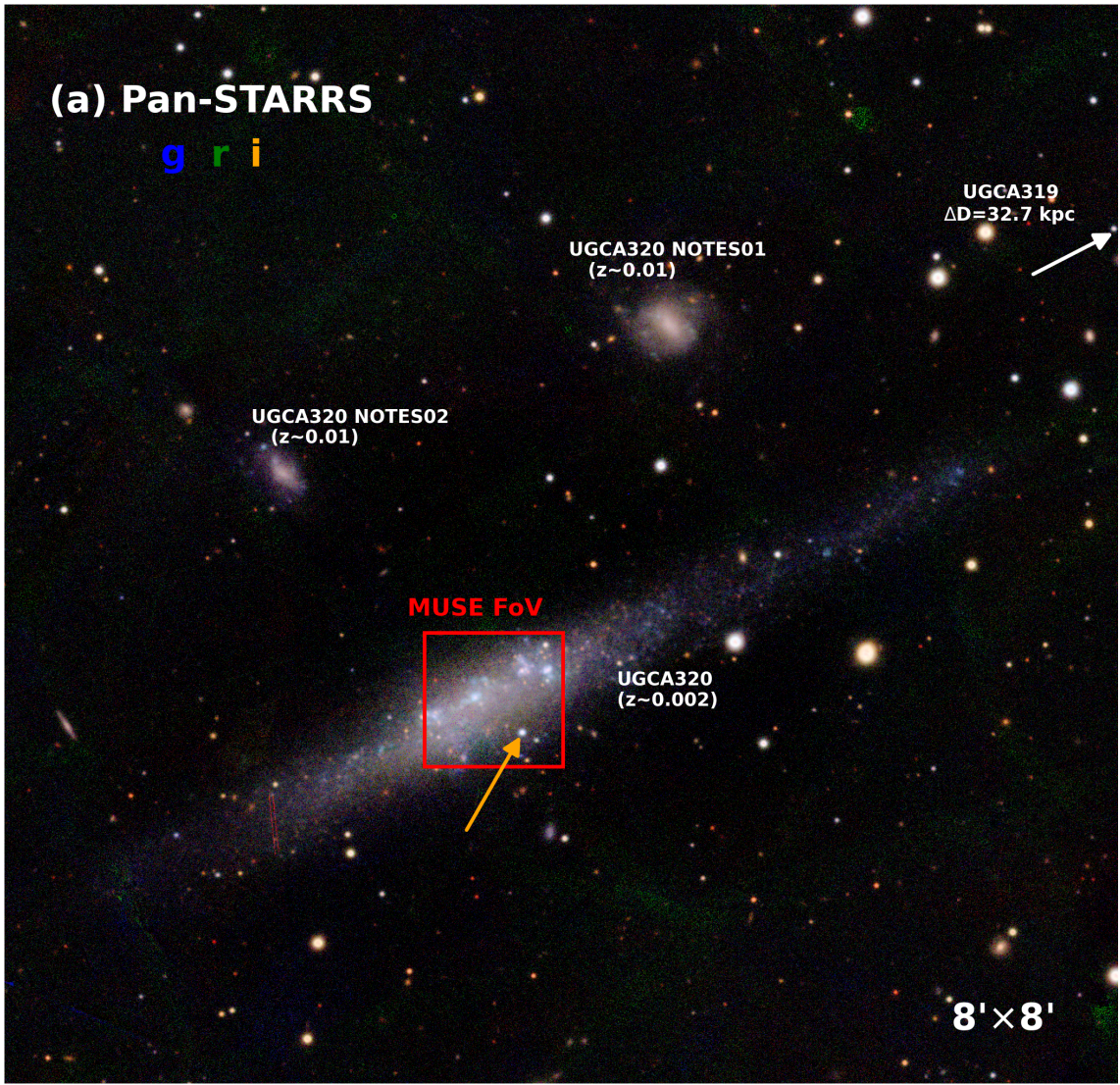


Figure 1 | HST and MUSE views pinpoint the wandering IMBH in UGCA 320, with velocity maps revealing its kinematic environment.

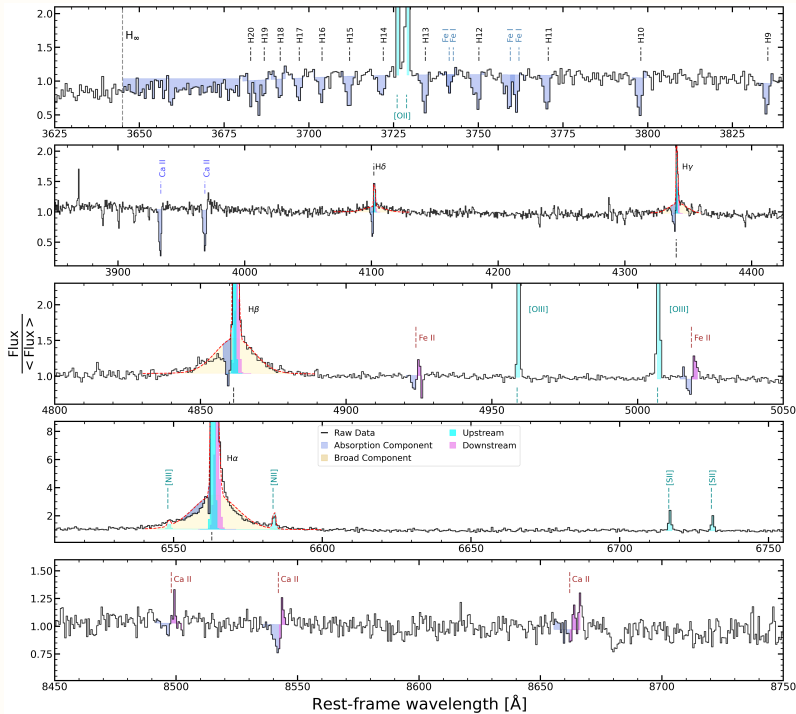


Figure 3 | X-Shooter spectrum decomposes the IMBH's gas into blueshifted upstream flow, redshifted wake, and broad AGN emission.